

House Price Prediction Using Linear Regression Lab Report 01

COURSE CODE: MDI3003

COURSE TITLE: Advanced Predictive

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Executive Summary

This study developed and evaluated regression models for predicting single-family home sale prices using the Ames Housing dataset. The engineering pipeline transitioned systematically from a single-feature Simple Linear Regression baseline up to Multiple Linear, Polynomial (with Ridge control), and Decision Tree architectures to isolate optimal predictive parameters.

The models were benchmarked using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). The Multiple Linear Regression model emerged as the champion configuration, achieving a high out-of-sample R^2 score of 0.8870 and a minimized RMSE of \$29,445.53. Clear indicators of target heteroscedasticity were monitored via residual analysis, providing a strict operational envelope for deployment safety parameters within the real estate valuation stream.

Business Problem Definition

- **Stakeholder Analysis:** Primary enterprise users include banking firms verifying mortgage risk, real-estate platforms automating property appraisal bounds, and developers filtering investment portfolios.
- **Operational Decision Context:** Automated value extraction to accelerate lending approvals, flag structural anomalies, and screen asset backings without human appraisal delays.
- **Cost of Model Errors:** Over-valuation distorts buyer affordability thresholds and exposes financial backers to systemic default risks. Conversely, severe under-valuation increases internal transactional friction and drops competitive bidding capabilities.
- **Success Threshold Criteria:** Model pipelines must confidently surpass a naive mean prediction baseline, limit train-test performance variance (overfitting), and maintain a stable error curve across residential price categories.

Dataset Profiling & Technical Description

The predictive pipeline was systematically trained and evaluated across the authoritative Ames Housing data matrix:

- **Observation Unit:** Individual residential property sales transactions located in Ames, Iowa.
- **Data Scale:** 1,460 standard entries feature extraction from the OpenML distribution (house_prices), partitioned into 1,168 Training rows (80%) and 292 Testing rows (20%).

- Target Metric and Unit: SalePrice — The actual absolute final transaction price of the home in USD.
- Primary Simple Feature: GrLivArea (Above grade ground living area square footage), chosen due to its high direct economic correlation to base real estate utility.
- Known Limitations: High-cardinality structural category flags introduce high dimensions post-encoding. Additionally, historical records reflect fixed baseline economic values that do not account for modern inflationary market drift.

Preprocessing & Leakage Control Strategy

To enforce academic integrity and prevent data leakage, data partitions were split prior to feature engineering. The preprocessing workflow was wrapped inside a leakage-safe ColumnTransformer pipeline:

- 1. Numerical Flow: Identified continuous data features (int64, float64) and resolved missing entries via a localized median imputation before applying a StandardScaler to align features to a mean of 0 and variance of 1.
- 2. Categorical Flow: Text fields were processed using a most_frequent token imputation strategy, followed by sparse expansion via OneHotEncoder to handle unknown categories gracefully during test inference.

Consolidated Result Tables &

Manual Templates Dataset

Summary Table

Dataset	Rows	predictors	Numeric	Categorical	Missing %	Target and Unit	Coverage
Dataset B (Ames)	1,460	79	36	43	5.86%	SalePrice (Absolute USD)	Ames, Iowa Residential Property Sales

Data-Quality Decisions Table

Issue	Evidence	Decision	Reason	Effect on Matrix Shape
Missing Values	Null entries observed in continuous/nominal matrices	Mixed Pipeline Median & Mode Imputation	Prevents matrix factorization breakdowns during model	Rows kept constant; dimensions preserved

			fitting stages	
Index Identifiers	Id column provides sequence numbers without predictive signal	Explicitly drop feature block from X	Stops models from memorizing row identity noise instead	Feature dimensions reduced by 1
Issue	Evidence	Decision	Reason	Effect on Matrix Shape
High Cardinality	Text metrics like Neighborhood hold multiple unique strings	Apply leak-safe OneHotEncoder	Translates text parameters safely into binary columns for linear processing	Expands column space post-transform

Final Test Comparison Table

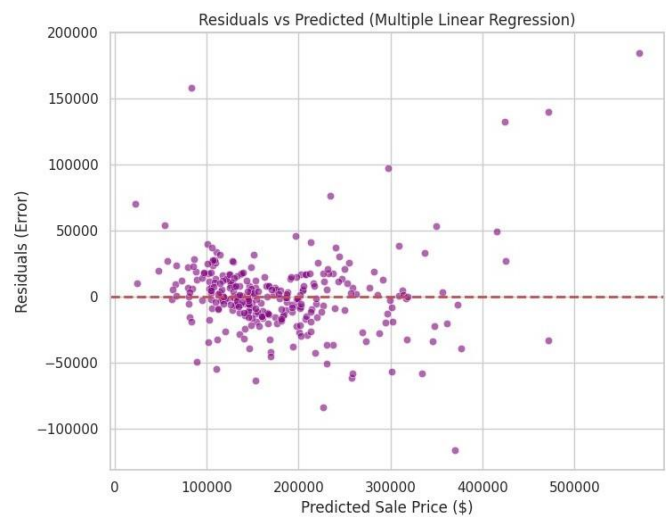
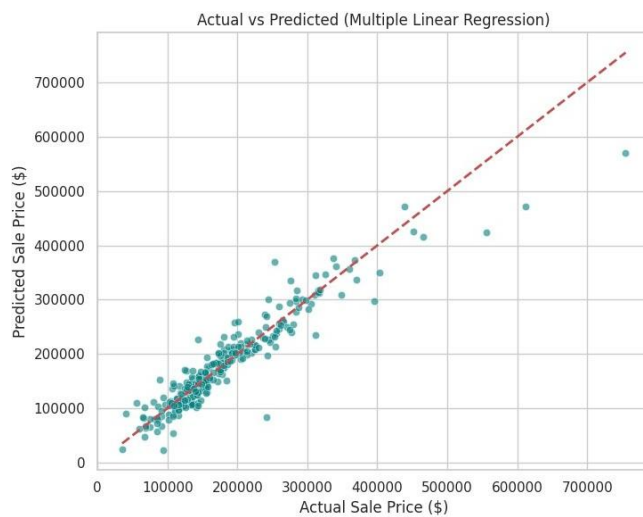
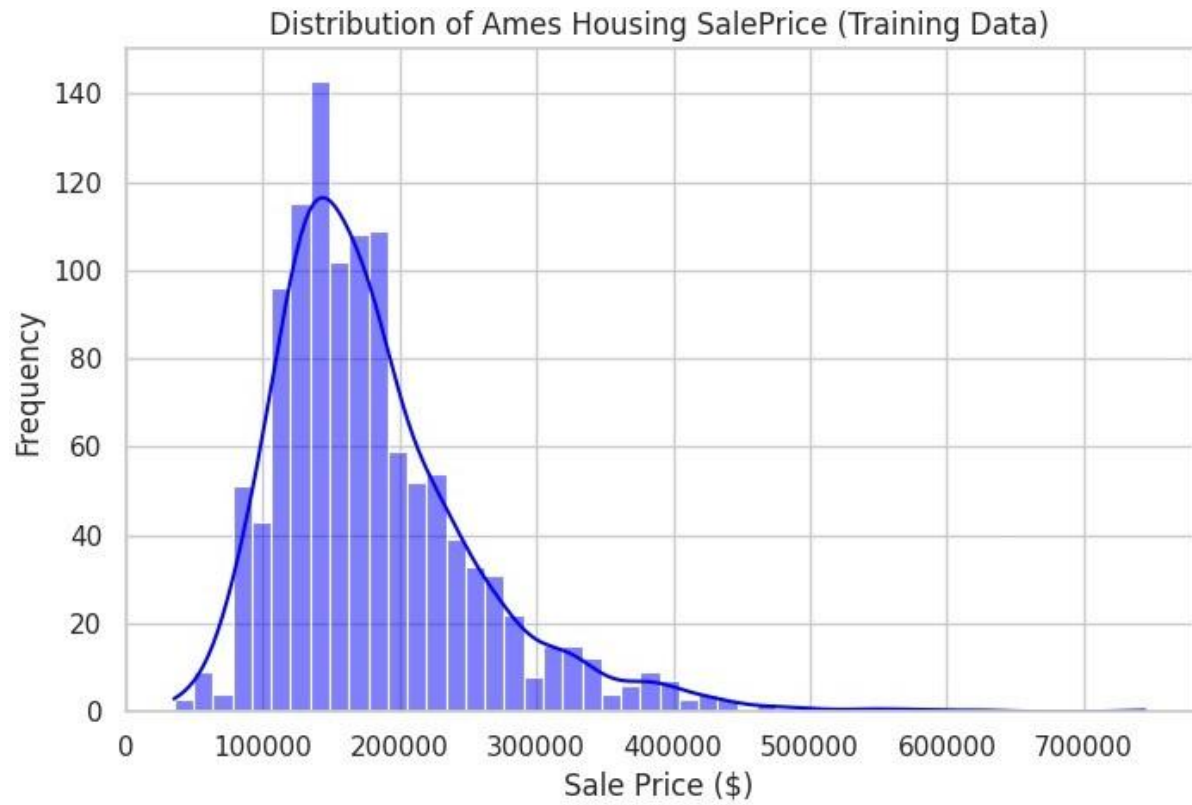
Model Configuration	Test MAE	Test RMSE	Test R2 Score	Generalization & Structural Notes
Simple Linear Baseline	\$38,341.20	\$58,471.76	0.5543	Low model capacity; relying solely on above-grade square footage underfits the true market pricing variance.

Multiple Linear Regression	\$18,362.37	\$29,445.53	0.8870	Relatively BEST configuration. Maximizes signal extraction across all numerical and encoded categorical variables.
Polynomial (Degree 2, Ridge)	\$22,652.52	\$34,723.64	0.8428	Captures interaction boundaries between top features (GrLivArea, OverallQual, TotalBsmtSF), but slightly overfits the tail.
Decision Tree ($d = 8$)	\$25,902.81	\$40,786.65	0.7831	Non-linear tree cuts allow automated interaction mapping, but rigid split limits weaken edge boundary smooth extrapolation.

Observation and Interpretation Table

Evidence	Observation	Interpretation	Action / Limitation
Target Distribution	Heavy right-skewed layout with a long trailing tail towards premium prices.	Mean squared error minimization fields focus heavily on high-end luxury property errors.	Integrate target log transformations ($y \rightarrow \log(y)$) in subsequent optimization passes.
Actual vs. Predicted	Strong linear alignment up to \$300k; prediction variance widens past that point.	The model performs cleanly on standard mid-tier assets but faces noise challenges in premium brackets.	Flag high-value predictions past \$350k for manual human appraisal review overrides.
Residual Plot	Error dispersion fan expands systematically as home value scales up.	Confirms distinct heteroscedasticity across higher asset pricing tiers.	Qualify risk statements; expand error tolerance boundaries for luxury listings.

Data Visual Plots



Final Model Recommendation & Synthesis

Select Multiple Linear Regression as the champion deployment architecture for this tabular matrix, as it provides the most optimal balance of predictive precision ($R^2 = 0.8870$, RMSE = \$29,445.53) and business interpretability. It yields a 49.6% reduction in root mean squared error compared to the single-feature baseline. While the Polynomial Ridge configuration captured localized high-value feature interaction trends effectively ($R^2 = 0.8428$), expanding columns without strict global regularized boundaries increased variance, confirming the stability of the baseline multivariate model. Before live deployment, check for spatial coordinate changes and ensure regular updates to handle changing market conditions.